

RESEARCH ARTICLE

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Key Points:

- Irrigation could lead to SW(GW) depletion depending on irrigation sources
- Irrigation water demand increases worldwide in response to climate change
- Accounting for source of irrigation is important in climate change assessment

Supporting Information:

- Supporting Information S1
- Figure S1
- Figure S2

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A modeling study of irrigation effects on global surface water and groundwater resources under a changing climate

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Abstract This study investigates the effects of irrigation on global water resources by performing and analyzing Community Land Model 4.0 (CLM4) simulations driven by downscaled/bias-corrected historical simulations and future projections from five General Circulation Models (GCMs). For each climate scenario, three sets of numerical experiments were performed: (1) a CTRL experiment in which all crops are assumed to be rainfed; (2) an IRRIG experiment in which the irrigation module is activated using surface water (SW) to feed irrigation; and (3) a PUMP experiment in which a groundwater pumping scheme coupled with the irrigation module is activated for conjunctive use of surface water and groundwater (GW) for irrigation. The parameters associated with irrigation and groundwater pumping are calibrated based on a global inventory of census-based water use compiled by the Food and Agricultural Organization (FAO). Our results suggest that irrigation could lead to two major effects: SW (GW) depletion in regions with irrigation primarily fed by SW (GW), respectively. Furthermore, irrigation depending primarily on SW tends to have larger impacts on low-flow than high-flow conditions, suggesting increased vulnerability to drought. By the end of the 21st century, combined effect of increased irrigation water demand and amplified temporal-spatial variability of water supply may lead to severe local water scarcity for irrigation. Regionally, irrigation has the potential to aggravate/alleviate climate-induced changes of SW/GW although such effects are negligible when averaged globally. Our study highlights the need to account for irrigation effects and sources in assessing regional climate change impacts.

1. Introduction

Freshwater resources are key to many aspects of the environment, society, and economy [Oki and Kanae, 2006]. A better understanding of the global water cycle allows for a better assessment of freshwater resources [Döll et al., 2012]. On the global scale, climate change has been shown to be the primary driver of hydrological cycle changes [Gedney et al., 2006; Taylor et al., 2013; Zhang et al., 2014]. However, the hydrological cycle is influenced not only by climate, but regulation by human activities such as irrigation and groundwater pumping is significant at the regional scale [Haddeland et al., 2006; Tang et al., 2007; Hanasaki et al., 2008; Ozdogan et al., 2010; Wada et al., 2010], which may also respond to climate [Green et al., 2011; Taylor et al., 2013] and affect the local and regional climate [Chen and Xie, 2010; Zou et al., 2015], leading to complex interactions. Broadly, climate impacts the hydrological cycle directly via water and energy exchanges [Elliott et al., 2014; Portmann et al., 2013; Piontek et al., 2014] and indirectly via changes in human activities that alter the hydrological cycle through water use [Taylor et al., 2013; Haddeland et al., 2014] and land use/cover change [Piao et al., 2007]. Specifically, around 70% of global freshwater in 2000 was withdrawn for irrigation, which accounted for 90% of consumptive water use [Döll, 2002]. Such large-scale redistributions of fresh water from rivers, lakes, reservoirs, and groundwater to agricultural regions have significantly perturbed the local/regional surface/subsurface hydrology [Rodell et al., 2009; Kustu et al., 2010; Famiglietti et al., 2011; Leng et al., 2014a; Sorooshian et al., 2014].

Climate change adaptation and mitigation in the water resources sector require an improved understanding on hydrologic responses to anthropogenic perturbations, including local water management [Taylor et al., 2013]. While many studies have evaluated individual impacts of climate change [e.g., Fowler et al., 2007; Wetherald, 2010], or irrigation/groundwater pumping on terrestrial hydrology at the regional scale [e.g., Ozdogan et al., 2010; Ferguson and Maxwell, 2012; Condon and Maxwell, 2014; Leng et al., 2014a], studies

investigating the joint effects of climate change and irrigation at the global scale are still limited. *Döll and Müller Schmied* [2012] investigated the impact of climate change on river flow regimes with considerations of human water use. Their results show that neglecting the effects of water withdrawals, climate change impact by the 2050s would be underestimated by at least 10% over ~5% of the global land area. Based on multiple land surface models, *Haddeland et al.* [2014] investigated the joint effects of climate change and irrigation on the global water resources and show that irrigation has comparable effects on runoff over the midlatitude arid regions as climate warming. *Forzieri et al.* [2014] showed that future climate change in Europe will lead to more severe conditions of low flow. This situation will be further aggravated by 10–30% in southern Europe while in some subregions in western, central, and eastern Europe, the climate-driven decrease of droughts may be reversed due to intensive water use. However, the effects of irrigation and climate change on the groundwater resource in addition to surface water have not been explicitly examined in these studies.

To date, few large-scale hydrological models have the ability to explicitly consider the effects of irrigation water use by source. For example, constrained by census data, *Döll et al.* [2012] estimated the water withdrawals from surface water and groundwater and their impacts on water storage variations globally using the WaterGAP model with a focus on the groundwater, in which the groundwater storage is represented as a linear reservoir, without considering the interactions between soil moisture and water table dynamics. *Wada et al.* [2014] integrated the global water demand model developed by *Wada et al.* [2011a, 2011b] with the global water balance model PCR-GLOBWB [*Wada et al.*, 2010] and simulated water withdrawal and consumptive water use from surface water and groundwater resources. Similar to *Döll et al.* [2012], the two-way interactions between soil moisture and groundwater were not explicitly considered in this model. Meanwhile, catchment-scale-integrated hydrological models are featured with more explicit and detailed representations of surface and subsurface dynamics. And the importance of irrigation management and groundwater pumping have also been increasingly recognized [e.g., *Faunt*, 2009; *Ferguson and Maxwell*, 2012; *Condon and Maxwell*, 2014]. Land surface models (LSMs) are an integral component of the General Circulation Models (GCMs) or Earth System Models (ESMs). Representing human impacts in LSMs is of great significance for investigating impacts of continental to global-scale environmental issues associated with climate change. However, few LSMs have the ability to account for human impacts such as irrigation [e.g., *Ozdogan et al.*, 2010; *Lo and Famiglietti*, 2013; *Sorooshian et al.*, 2014] and groundwater pumping. Recently, *Pokhrel et al.* [2014] developed a groundwater pumping module within the framework of a global LSM, in which the deficits between the total demand and the surface water availability are met by withdrawing water from the aquifers. By setting the model soil moisture at a fixed percentage of field capacity, however, the simulated water use from surface water and groundwater was not effectively constrained by observations that are usually available at finer scales. *Zou et al.* [2014] incorporated a groundwater pumping scheme into the Community Land Model (CLM) version 3.5, with the irrigation water demand prescribed as a constant rate without accounting for the interannual and seasonal variability. *Leng et al.* [2014a] implemented a groundwater pumping scheme into CLM4 for the United States and partitioned the calibrated irrigation water demand, which is estimated dynamically based on soil water balance, into surface water and groundwater withdrawals based on the county-scale agricultural census data. Their results showed that groundwater-fed irrigation could lead to depletion of groundwater resources in regions with shallow water table depths and low recharge rates, which is consistent with observations from groundwater monitoring wells from USGS in regions such as the Ogallala aquifer.

Because water extraction from aquifer storage for irrigation usually occurs during the growing season, it is hypothesized that groundwater-fed irrigation could aggravate the impact of climate change on below-normal conditions of groundwater storage. *Ferrant et al.* [2014] moved a step forward by integrating impacts of climate change on groundwater withdrawals and their combined effects on the evolution of groundwater resource, but limited their study to the crystalline aquifers in South India. Up to now, investigations of the impacts of irrigation water use in the context of climate change on surface and subsurface water resources were rare at the global scale. Understanding how human water use would respond to global warming and its combined effects on the hydrologic cycle is important for better designing mitigation and adaption strategies to the global change in the future. In this study, we take advantage of the global inventory data set on withdrawals from surface water and groundwater at the resolution of 0.5° developed by *Siebert et al.* [2010]. This data set enables us to quantitatively explore the combined effects of

climate change and irrigation withdrawals from surface water and groundwater on the global surface and subsurface hydrology at a relatively fine scale. This study aims to (1) extend and calibrate/validate the groundwater pumping scheme coupled with CLM4 as developed by *Leng et al.* [2014a] at the global scale and (2) investigate the combined effects of climate change and irrigation on both surface and subsurface hydrology in the 21st century. This paper is organized as follows: in section 2, the data sets, model, and experimental design are described. In section 3, results on model validation and climate change and irrigation impacts on global surface water/groundwater are presented. In section 4, the limitations of this study are discussed. Section 5 summarizes the findings of this study.

2. Data and Methodology

2.1. Model Descriptions

CLM4 is the land surface component of and can be used in offline or coupled modes within the framework of the Community Earth System Model (CESM) [*Gent et al.*, 2010; *Lawrence et al.*, 2011] or the Weather Research and Forecasting (WRF) model [*Leung et al.*, 2006; *Ke et al.*, 2012; *Gao et al.*, 2014]. CLM4 utilizes a subgrid hierarchy with multiple soil columns coexisting in a grid cell and allow multiple plant functional types (PFTs) to exist in one soil column in which the dynamics for soil water, soil organic carbon, litter, etc. are represented [*Lawrence et al.*, 2011]. CLM4 includes several advances over its prior versions such as improved canopy hydrology, snow hydrology, soil hydrology, and evapotranspiration [*Oleson et al.*, 2010]. The model has been comprehensively evaluated in previous studies [e.g., *Li et al.*, 2011; *Hou et al.*, 2012; *Huang et al.*, 2013] and widely used in investigating the spatial-temporal variability of the hydrological cycle at regional and global scales [e.g., *Liu and Xie*, 2013; *Shi et al.*, 2013].

A soil moisture dependent irrigation scheme has been incorporated into CLM4 recently [*Leng et al.*, 2013] and has been applied successfully to study the effects of irrigation in regions such as the U.S. and China [e.g., *Leng et al.*, 2013, 2014a, 2014b; *Leng and Tang*, 2014]. Irrigation modifies the water and energy budgets (e.g., soil moisture, evapotranspiration, and runoff) in the simulations after irrigation water was applied to the soil column. Given that only C3 generic crop is represented in CLM4 and data on irrigation methods are not available at the global scale, the characteristics of different crop types, crop growth, and irrigation methods are not considered in this study. Details of the irrigation scheme can be found in *Leng et al.* [2013]. Briefly, the cropland area of each grid cell is divided into irrigated and unirrigated fractions based on a data set of areas equipped for irrigation [*Siebert et al.*, 2005; *Ozdogan et al.*, 2008]. For irrigated croplands, irrigation is required when the crop leaf area > 0, and $\beta_t < 1$ (i.e., water is limiting photosynthesis), and the irrigation water demand (IWD) is calculated as the total soil moisture deficit in the root zone based on a check at 6:00 A.M. on each day. That is, for each soil layer, the deficit between the current soil moisture content and the target soil moisture content was calculated and aggregated as the IWD. The target soil moisture content of each soil layer ($W_{target,jr}$ kg m⁻²) is given by

$$W_{target,j} = F_{irrig} \times W_{sat,j} + (1 - F_{irrig}) \times W_{o,j} \quad (1)$$

where $W_{o,j}$ is the minimum soil moisture content below which water stress for crop growth will occur; $W_{sat,j}$ is the soil moisture content at saturation in that layer; F_{irrig} is a weighting parameter that varies between 0 and 1, corresponding to the soil moisture target that is just enough to result in no water stress for crops or at full soil saturation. Since some assumptions (see section 4) have to be made due to limitations associated with data availability and irrigation representations in CLM4, F_{irrig} should be treated as a conceptual parameter used to calibrate simulated irrigation amounts against the reference data.

The estimated irrigation water demand (IWD) can be met by withdrawing water from local surface water [*Leng et al.*, 2013] and/or groundwater [*Leng et al.*, 2014a, 2014b] by partitioning the total irrigation withdrawals into surface and subsurface withdrawals. When water is pumped out from the aquifer storage, the corresponding storage terms and water table are updated. This way, the effects of groundwater withdrawals on groundwater dynamics could be explicitly examined. Details for the groundwater pumping scheme should be referred to *Leng et al.* [2014a]. The irrigation water withdrawal is then applied evenly to the soil beneath the irrigated crops during 6:00–10:00 A.M. on each day of the growing season. When local surface water and groundwater are not enough for irrigation at the time step, water withdrawals from other grids within the basin are allowed [*Leng et al.*, 2014a]. Even with this additional water resource, the irrigation

demand may still not be met in typical dry seasons since dynamic reservoir operations are not represented. This gap is then filled by mimicking the process of water diversion from outside the river basin, and the surface water variables are set to have negative values to represent the deficit. This way, a negative value of surface water indicates the deficit of irrigation demand and basin-averaged renewable water resource, which should be met by water diversion from outside the river basin.

2.2. Irrigation Water Use From Surface Water and Groundwater

Siebert *et al.* [2010] provided a global inventory for irrigation water use, with the sources of irrigation water indicated. This global inventory, which is mainly based on national census reports from online databases or from the FAO-AQUASTAT library, includes the area equipped for irrigation with groundwater, surface water, and water derived from nonconventional sources for 15,038 national and subnational administrative units. By linking these statistics to the Global Crop Water Model [Siebert and Döll, 2010], consumptive irrigation water use from surface water and groundwater has been estimated at the resolution of 0.5° globally. In this study, the total consumptive irrigation water use data are taken as the reference (denoted as irr_{total}) for our model calibration. The total consumptive irrigation water use and the consumptive irrigation water use from groundwater (denoted as irr_{grd}) are used to calculate the key parameters for the groundwater pumping scheme:

$$F_{grd} = \frac{irr_{grd}}{irr_{total}} \quad (2)$$

$$F_{surf} = 1 - F_{grd} \quad (3)$$

where F_{grd} and F_{surf} represent the ratios of irrigation water use satisfied by groundwater and surface water, respectively. The r_{total} and r_{grd} are estimated for each administrative unit and remain constant temporally throughout the simulations to be discussed in detail in section 4.

2.3. Observed Historical Climate and Climate Projections

The Water and Global Change (WATCH) Forcing Data (WFD) [Weedon *et al.*, 2011] are used as the observed historical climate. The WFD are produced by combining the ERA-40 daily data, the 40 years reanalysis of the European Centre for Medium-Range Weather Forecasts (ECMWF), and the Climate Research Unit TS2.1 data set (CRU) for the overlapping periods of 1960–1999. The WFD data set features a combination of daily variability from ERA-40 with the monthly mean characteristics derived from the CRU and GPCC data sets.

Climate projections from five GCMs from the Coupled Model Intercomparison Project Phase 5 (CMIP5) archive are obtained. The climate projections under four Representative Concentration Pathways (RCPs), which are provided within the framework of Inter-Sectoral Impact Model Intercomparison Project (ISI-MIP) [Warszawski *et al.*, 2013], were bias-corrected against the WFD for the overlapping period 1960–1999. After bias correction, the climate projections represent well the observed climate for the reference period in terms of mean and variability [Piani *et al.*, 2010; Hempel *et al.*, 2013]. Up to now, projections from five GCMs are provided by ISI-MIP. Given that the goal of this study is not to provide comprehensive assessment with associated uncertainty, but rather to provide illustrative examples of possible changes and common behaviors across the projected future climate change, the relatively small ensemble in ISI-MIP is sufficient for our purpose. This data set has been used for quantifying the impacts of climate change and climate variability on the hydrological cycle at regional and global scales [e.g., Portmann *et al.*, 2013; Wada *et al.*, 2013; Piontek *et al.*, 2014; Leng *et al.*, 2014c, 2015]. Details on the five GCMs can be found in Table 1. Among the four emission scenarios, we selected the RCP8.5 scenario because it represents the highest change in greenhouse gas concentration (>1370 ppm CO_2 equivalent) and radiative forcing (up to 8.5 W m^{-2}) [Moss *et al.*, 2010; Rogelj *et al.*, 2012], therefore, the largest possible impacts of climate change [e.g., Piontek *et al.*, 2014; Elliott *et al.*, 2014; Scheweet *et al.*, 2014]. The climate data include daily precipitation, air temperature, wind speed, surface radiations, air pressure, and specific humidity at $0.5^\circ \times 0.5^\circ$ spatial resolution from 1950 to 2099 [Hagemann *et al.*, 2013]. Temporal disaggregation is performed to produce subdaily climate forcing for CLM4 as described by Leng and Tang [2014].

2.4. Experimental Design

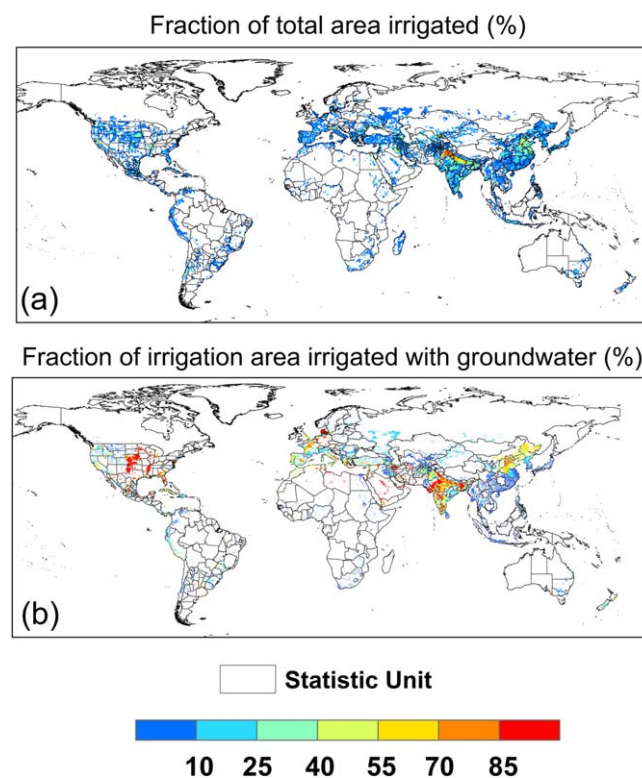
For irrigation modeling studies, accurate estimation of irrigation water amounts is crucial [Sacks *et al.*, 2009; Sorooshian *et al.*, 2012; Leng *et al.*, 2013, 2014a]. Hence, the model was first calibrated to match the

Table 1. Descriptions of the Five GCMs Used in This Study

ID	GCM Name	Institute Acronyms	Institute Full Name
1	HadGEM2-ES	MOHC [additional realizations by INPE]	Met Office Hadley Centre and Instituto Nacional de Pesquisas Espaciais
2	IPSL-CM5A-LR	IPSL	Institut Pierre-Simon Laplace
3	GFDL-ESM2M	NOAA GFDL	NOAA Geophysical Fluid Dynamics Laboratory
4	MIROC-ESM-CHEM	MIROC	Japan Agency for Marine-Earth Science and Technology, Atmosphere and Ocean Research Institute [The University of Tokyo], and National Institute for Environmental Studies
5	NorESM1-M	NCC	Norwegian Climate Centre

observed irrigation water use at the administrative level, which was done manually following the method by *Leng et al.* [2013], in which the value of F_{irrig} was perturbed between [0,1] at a regular interval of 0.1. The optimal value of F_{irrig} was selected for each administrative region by comparing the simulated irrigation amounts aggregated in that region against the corresponding observations (i.e., the consumptive irrigation water use from *Siebert et al.* [2010]). It should be noted that the boundaries of the administrative units used in this study are slightly different from that archived by *Siebert et al.* [2010] (i.e., shown in *Siebert et al.* [2010, Figure 1]) due to the limitation of access to the map (S. Siebert, personal communication, 2013). All simulations with irrigation turned on in this study adopted the calibrated parameter values (i.e., F_{irrig}) in modeling IWD.

Descriptions of the experiments conducted in this study can be found in Table 2. Specifically, based on the WFD data, three simulations were conducted for the period 1951–2000: CTRL, IRRIG, and PUMP. The CTRL assumes a “natural” condition in which irrigation is not considered. Both IRRIG and PUMP are the same as CTRL but with the irrigation module or groundwater pumping coupled to the irrigation module activated, respectively. To be consistent with the ISI-MIP modeling framework, the first 20 years of simulations were discarded and the following 30 years (i.e., 1971–2000) of results were used in this study. In the IRRIG scenario, IWD was met only by withdrawing water from SW (i.e., local total runoff within the grid cell) while in the PUMP scenario, which is more representative of the real world than IRRIG, IWD was met by withdrawing water from SW and GW (i.e., aquifer storage (mm)) concurrently based on the ratio (i.e., r_{grd} and r_{surr}) archived by *Siebert et al.* [2010]. By comparing the results from IRRIG and PUMP, the effect of accounting for irrigation sources in climate change impact assessments is examined. It should be noted that IWD is fully met in both IRRIG and PUMP by setting the model to withdraw SW from other remote areas within the specific river basin when local water resources (SW in IRRIG, SW and GW in PUMP) is not enough for irrigation. When water resources for the whole basin is still not enough for irrigation, water is assumed to come from other river basins by setting the SW to be negative to denote water deficit.


Figure 1. Spatial distribution of the fraction of irrigated area and irrigated area with ground by *Siebert et al.* [2010].

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Table 2. Descriptions of Simulations in This Study

Name	Climate Data Set [Simulation Period]	Irrigation	Calibrated F _{irrig}	F _{surfl} / F _{grd}
CTRL	WATCH forcing	No		
IRRIG	[1951–2001]	Yes	Yes	No
PUMP		Yes	Yes	Yes
CTRL _{GCMi}	The <i>i</i> th GCM forcing	No		
IRRIG _{GCMi}	[1951–2001]	Yes	Yes	No
PUMP _{GCMi}		Yes	Yes	Yes

Based on the downscaled and bias-corrected climate projections from ISI-MIP, three simulations were conducted for each GCM over the period of 1951–2009, i.e., CTRL_{GCMi}, IRRIG_{GCMi}, and PUMP_{GCMi}, where *i* indicates the *i*th GCM. Hence, there are 15 simulations (3 × 5 GCMs) in total. The climate-induced impacts were calculated as the difference between the simulated

variables for the future period (i.e., 2071–2099) and historical period (1971–2000) from CTRL_{GCMi}. The combined effects of climate change and irrigation were represented by calculating the difference between IRRIG_{GCMi} (or PUMP_{GCMi}) for the future period and CTRL_{GCMi} for the historical period. To facilitate the analysis, we also calculate the multimodel ensemble mean, denoted as CTRL_{ISI-MIP}, IRRIG_{ISI-MIP}, and PUMP_{ISI-MIP}, respectively.

Land cover composition and land surface parameters such as PFTs, Leaf Area Index (LAI), Stem Area Index (SAI) of the grid cells were derived from the standard CLM input data sets provided by NCAR accompanying the CLM4 release. That is, lake and wetland data were from Global Hydrographic Data in 1991 [Cogley, 1991]; PFTs fractional cover data were derived using a combination of the MODIS Vegetation Continuous Field (VCF) and MODIS land cover product around 2000, and 1992–1993 AVHRR Continuous Field Tree Cover Project data [Lawrence and Chase, 2007; Lawrence et al., 2011]; phenological parameters such as LAI, SAI were derived by averaging high-quality MODIS LAI data during 2001–2003 onto coarser resolution (0.5°) grid cells, and partitioning the averaged LAI for each PFT. The irrigation fraction map was obtained from the widely used data set representing the fraction of irrigated area around year 2000 by Siebert et al. [2005]. All gridded input data sets were interpolated to 0.5° resolution for use by CLM4. Before the simulations, an off-line spin-up was performed by cycling the meteorological forcing for 20 cycles using the CLM_CRTL setup until equilibrium conditions were achieved for all state variables including soil moisture, temperature, and groundwater table depth. The resulting state variables were saved and used to initialize other simulations.

3. Results

3.1. Key Parameters and Model Calibration

Figure 1 shows the spatial distributions of the percentages of total irrigated areas and ratio of groundwater-fed irrigation (i.e., F_{grd}). The map of F_{grd} was generated by combining the subnational groundwater inventory with the percentage of total irrigated area [Siebert et al., 2010]. About 70% of the irrigated area is located in Asian countries, followed by North America, Europe, South America, Africa, and Oceania. At the country scale, the largest extent of irrigated areas is reported for China, India, the U.S., and Pakistan, which occupy about 56% of the global irrigated area. The largest irrigated areas fed by groundwater are located in India, followed by China, the U.S., Pakistan, Islamic Republic of Iran, Bangladesh, Mexico, Saudi Arabia, Turkey, Spain, Italy, and France (Figure 1b). At the regional scale, the highest density of irrigated area fed by groundwater was identified for the Ganges and the upper Indus watersheds, Bangladesh, the North China Plain, the High Plains aquifer and the alluvial aquifer along the Mississippi River in the U.S.

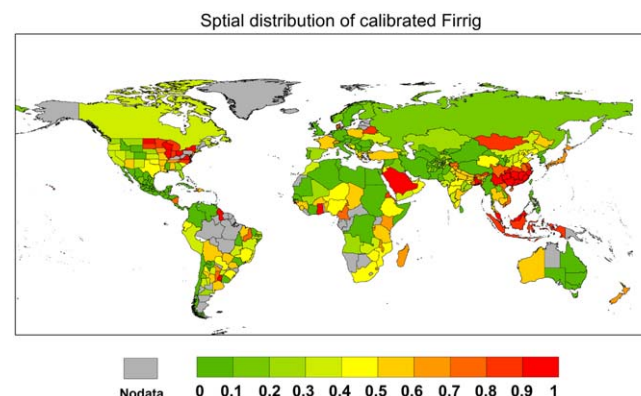

Figure 2. Spatial distribution of the calibrated F_{irrig} at the administrative scale.

Figure 2 shows the spatial distribution of the calibrated F_{irrig} values for each administrative region. High values of F_{irrig} up to 0.8 are distributed over South China, Northwest U.S., and

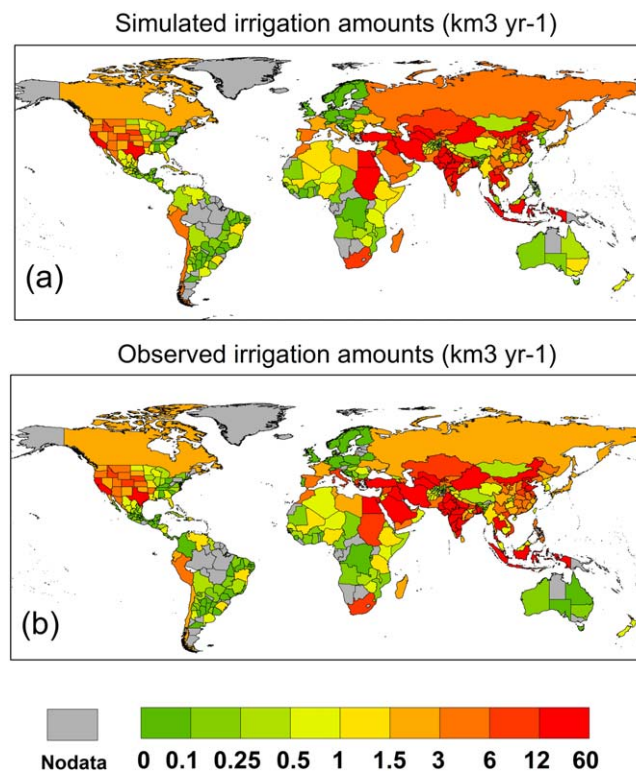


Figure 3. Spatial distributions of annual irrigation amounts ($\text{km}^3 \text{yr}^{-1}$) simulated by (a) CLM4 after calibration and (b) observation data set by Siebert *et al.* [2010].

River and Haihe River), and the United States (Colorado River and Mississippi River), are selected to illustrate the role of irrigation on the changes in surface water and groundwater in the following analysis. Figure 5 shows the monthly climatology of runoff simulated by the CTRL, IRRIG, PUMP, and Global Runoff Data Centre (GRDC) composite runoff data [Fekete *et al.*, 2002] for the six study regions and global average. Generally, the model captured the seasonal variability of runoff for the six river basins and the whole world. When the irrigation scheme is activated, smaller bias between the simulations and observation are found compared to the control scenario, especially for basins with intense irrigation activities such as Yellow River, Haihe River, and Colorado River basins. However, there is still large discrepancy in terms of the seasonal

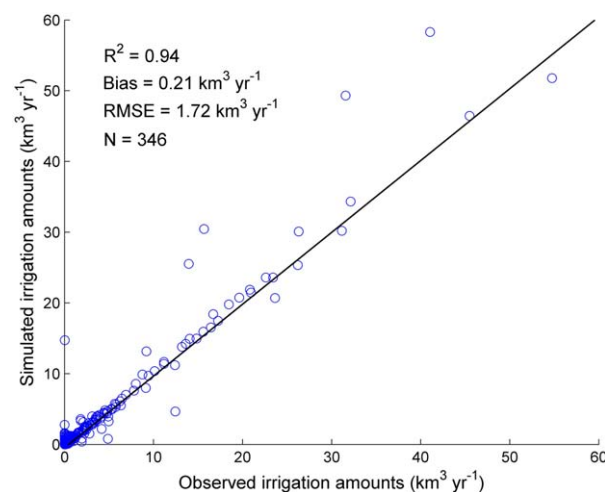


Figure 4. Scatter plot of simulated irrigation amounts after calibration versus observation data set at the administrative level.

Mongolia while values of F_{irrig} lower than 0.5 are found in most dominant irrigated regions such as North China Plain, Northern India, and California. Comparison of annual simulated IWD ($\text{km}^3 \text{yr}^{-1}$) using the calibrated F_{irrig} values with that by Siebert *et al.* [2010] indicates that the geographical features of irrigation amounts are captured reasonably well after calibration (Figure 3). Relatively large irrigation amounts higher than $3 \text{ km}^3 \text{yr}^{-1}$ are found in large irrigated district such as North China plain, Northern India, Pakistan, California, and Southern Great Plain of the U.S. The statistics in Figure 4 show that the mean bias and RSME was -0.21 and $1.72 \text{ km}^3 \text{yr}^{-1}$, respectively.

3.2. Historical Impacts of Climate Change and Irrigation

It is expected that irrigation impacts on water resources are more obvious at the regional scale. Therefore, six focus regions, including the agriculturally important portions of India (Indus River and Ganges River), China (Yellow River and Haihe River), and the United States (Colorado River and Mississippi River), are selected to illustrate the role of irrigation on the changes in surface water and groundwater in the following analysis. Figure 5 shows the monthly climatology of runoff simulated by the CTRL, IRRIG, PUMP, and Global Runoff Data Centre (GRDC) composite runoff data [Fekete *et al.*, 2002] for the six study regions and global average. Generally, the model captured the seasonal variability of runoff for the six river basins and the whole world. When the irrigation scheme is activated, smaller bias between the simulations and observation are found compared to the control scenario, especially for basins with intense irrigation activities such as Yellow River, Haihe River, and Colorado River basins. However, there is still large discrepancy in terms of the seasonal amplitude of runoff, as a result of combined effects of model assumptions/parameterizations, resolutions as well as uncertainties in meteorological forcing data set (e.g., precipitation, temperature, and radiations). It should also be noted that the GRDC runoff climatology is a combination of model outputs and observations. Hence, caution should be taken when interpreting the results although it was constrained by discharge measurements. To decompose and diagnose the uncertainty sources from both simulations and observation requires further efforts. In this study, we will focus on the effects of irrigation withdrawals from different sources on natural runoff seasonality under a changing climate as simulated by the model. Therefore, the ability of the model to reproduce observed runoff and its

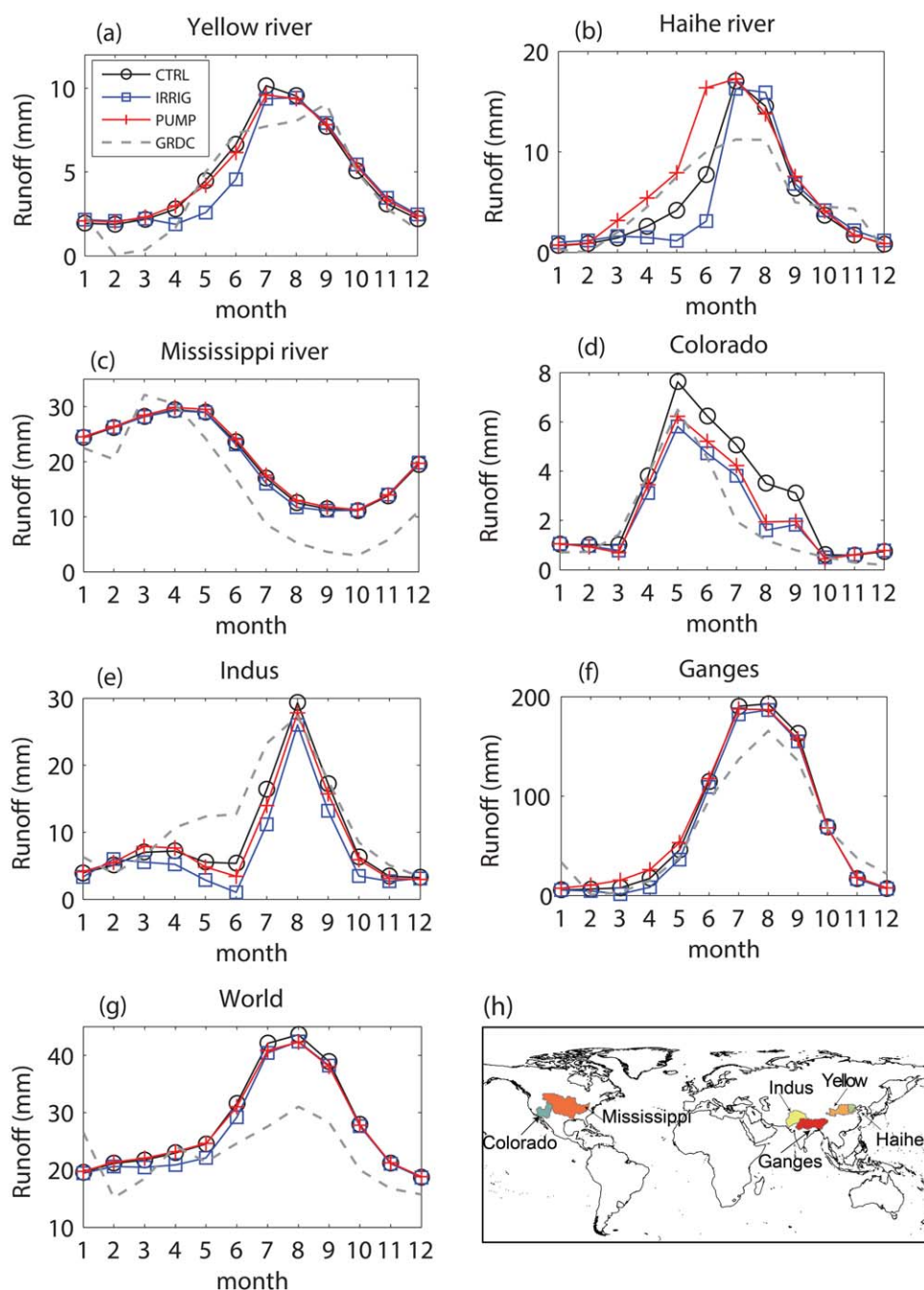


Figure 5. Comparison of monthly runoff climatology simulated by the CTRL, IRRIG, PUMP and the observation based GRDC for the region and global averages for 1986–1995.

seasonal patterns should be adequate for our following analysis on irrigation effects in the historic and future periods.

Figure 6 shows the cumulative frequency distributions (CDF) of regionally averaged SW (i.e., total runoff) constructed based on the simulations by the CTRL, IRRIG, and PUMP for the six study regions and the whole world. In the six river basins, irrigation leads to decrease of SW throughout the CDF curve in general as estimated by IRRIG (compared to CTRL). Importantly, irrigation tends to have much larger impacts on low-flow conditions as evident by the larger changes in the lower percentiles of SW compared to that in the high percentile values. For example, the magnitude of decrease in the tenth percentile of SW generally doubles that in the mean SW as estimated by IRRIG (compared to CTRL) for all six river basins (Table 3). Our results

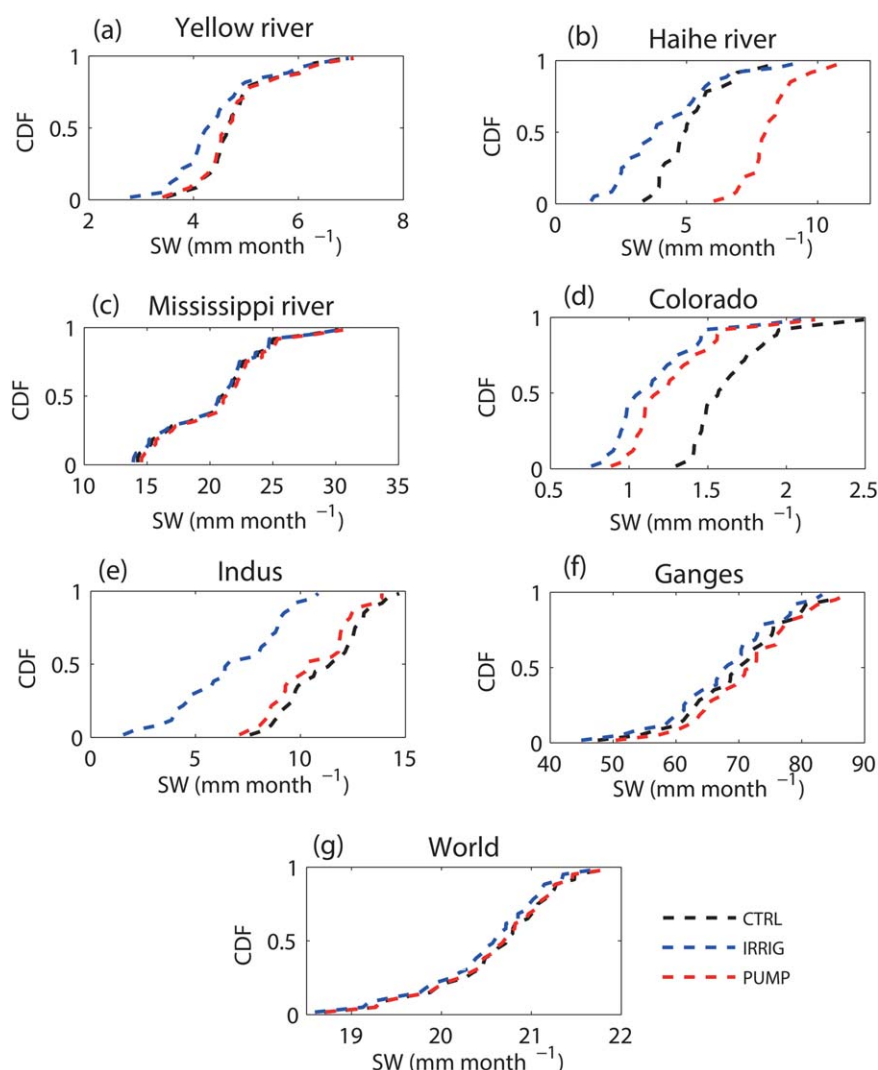


Figure 6. Cumulative distribution function (CDF) for surface water (mm month^{-1}) constructed based on the simulations by the CTRL (black line), IRRIG (blue line), and PUMP (red line) for the historical period 1971–2000.

indicate that intensive water use can further exacerbate low-flow conditions in watersheds, with risks of increased vulnerability to drought. In the PUMP scenario, part of IWD is met by withdrawing water from aquifer storage, allowing possible increase of SW by irrigation return flow. This is demonstrated by comparing the changes in SW in IRRIG and PUMP, which showed much smaller decrease of SW in PUMP than IRRIG. In river basins relying heavily on groundwater for irrigation, SW could even increase (compared to CTRL) due to the large amounts of water pumped out from aquifer storage (Table 3), which is applied to the soil surface and potentially become surface return flow. In extreme scenarios in Haihe River which is featured with relatively dry land (i.e., low SW in naturalized conditions, see Figure 6b), irrigation can lead to significant percentage increase up to 50% in SW if the irrigation water was mainly from aquifer storage (i.e., $r_{\text{grd}} > 70\%$ in most irrigated grids).

Similarly, irrigation could lead to two major opposing effects on GW. That is, irrigation could result in groundwater accumulation in regions where irrigation is primarily fed by surface water, and groundwater depletion in regions where irrigation is primarily fed by groundwater. For example, irrigation leads to general increases in GW if all IWD is met by SW as in the IRRIG scenario (Figure 7) due to the additional recharge as a result of irrigation and in turn percolation of the irrigated water through the vadose zone. In the PUMP scenario, on the other hand, irrigation could have mixed impacts on GW depending on the proportion of groundwater withdrawals, as well as the dynamics and interacting effects between surface and subsurface

Table 3. Statistics for the Changes (%) in Mean Monthly SW and Tenth Monthly SW for 1971–2000 by IRRIG and PUMP Compared to CTRL for the Six River Basins and the Whole World

	IRRIG-CTRL (%)	PUMP-CTRL (%)
<i>Yellow River</i>		
Mean monthly SW	−7.23	−0.19
Tenth monthly SW	−14.25	−3.31
<i>Haihe River</i>		
Mean monthly SW	−18.79	58.40
Tenth monthly SW	−42.71	80.74
<i>Mississippi River</i>		
Mean monthly SW	−0.75	1.63
Tenth monthly SW	−1.37	2.54
<i>Colorado River</i>		
Mean monthly SW	−29.18	−22.44
Tenth monthly SW	−37.88	−29.04
<i>Indus River</i>		
Mean monthly SW	−39.88	−5.39
Tenth monthly SW	−59.07	−5.79
<i>Ganges River</i>		
Mean monthly SW	−2.92	2.41
Tenth monthly SW	−4.27	4.11
<i>World</i>		
Mean monthly SW	−0.61	−0.08
Tenth monthly SW	−0.56	0.04

hydrology. For example, in the PUMP scenario, regions such as Yellow River and Colorado River with irrigation met primarily by SW, irrigation resulted in increases in GW similar to the IRRIG scenario, but at smaller magnitudes (Figures 7a and 7d). In extreme cases, intensive groundwater pumping for irrigation could lead to imbalance conditions (i.e., GW abstractions > GW recharge), depleting existing groundwater storage (e.g., in Haihe River and Indus River, see Figures 7b and 7e). For the whole world or large river basins such as Mississippi, the impacts of irrigation on the regional mean SW and GW are negligible (Figures 6g and 6g), further confirming the localized effects of irrigation as found by previous studies [e.g., Haddeland *et al.*, 2006; Ozdogan *et al.*, 2010; Leng *et al.*, 2014a].

3.3. Projected Changes in IWD and Water Scarcity

After bias correction, the climate projections used in this study were comprehensively validated against observations and found to reproduce well the

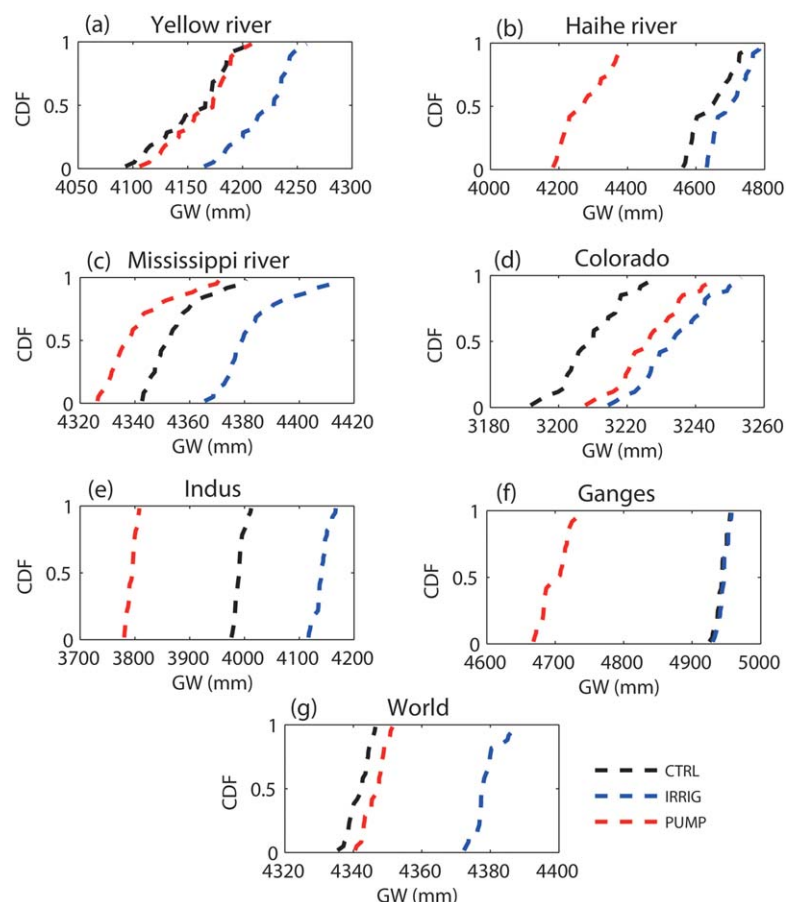


Figure 7. Same as Figure 6 but for groundwater (GW, mm).

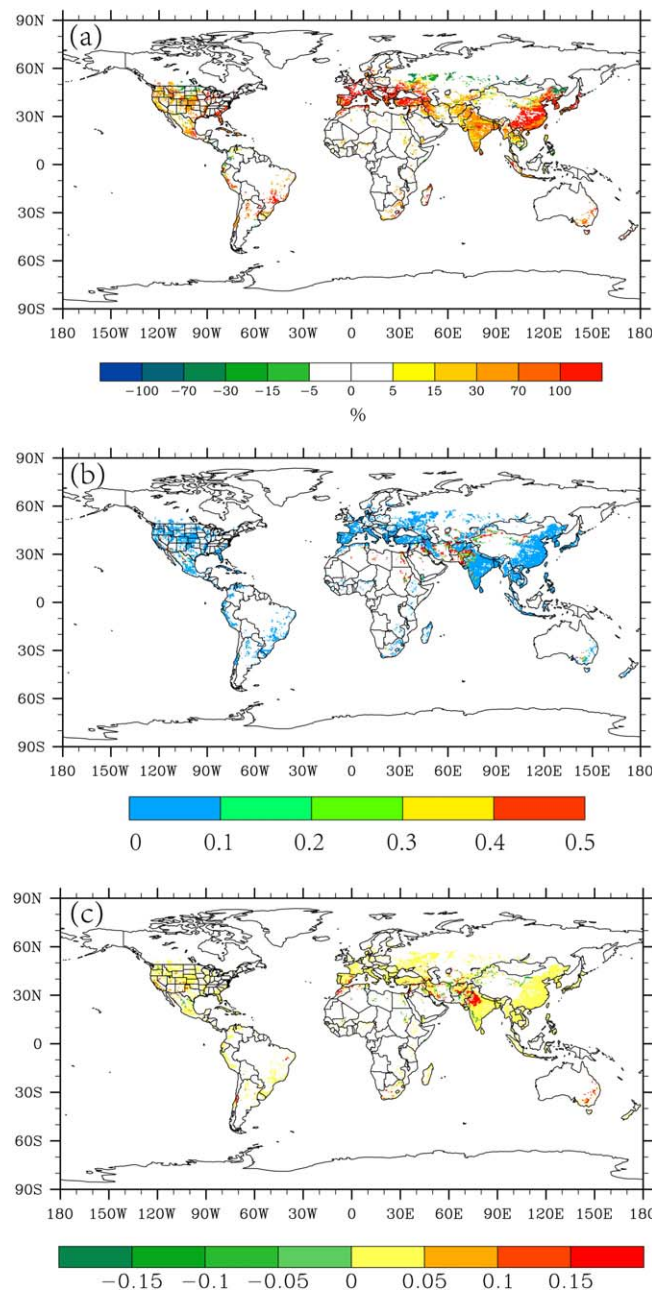


Figure 8. (a) Relative changes in irrigation amounts by 2070–2099 relative to 1971–2000, (b) spatial distribution of local water scarcity for irrigation (LWSI) for 1971–2000, and (c) changes in LWSI by 2070–2099 relative to 1971–2000.

than 50% is found over major portions of irrigated areas across Northern India, South Mexico, and Central United States. Over South China, Southeast United States, and Mediterranean regions, much larger increase in IWDs (i.e., over 100%) is projected.

In some cases, available local SW and GW combined is not enough for meeting IWD, leading to water scarcity. Figures 8b and 8c show the spatial distribution of local water scarcity for irrigation (LWSI, defined here as the ratio of water deficit divided by the water demand for irrigation) over the historical period and changes by the end of the 21st century, respectively. IWDs larger than the total local renewable water resources (i.e., SW + GW) are found for the irrigated areas across the world, indicating the requirement for non-local water resources (redistribution of water within the basin and/or water diversions from outside the river basin). In extreme cases, for irrigated regions over California in the U.S., Northern India, Pakistan, and

observed climate in terms of mean and variability [Piani *et al.*, 2010; Hempel *et al.*, 2013]. Since the focus of this study is on irrigation, it is essential to examine and validate the performance of IWD estimates over the historical period when CLM is driven by bias-corrected climate simulations. Comparison of the long-term mean IWD modeled using the WFD and adjusted climate by GCMs for the period 1971–2000 shows similar spatial structures and magnitudes (supporting information Figure S1). From the scatter plot of the IWD estimates for each administrative unit (supporting information Figure S2), it is found that the simulated IWDs driven by bias-corrected historical climate from the GCMs matches that driven by the observed climate very well ($R^2 = 0.95$). Thus, the modeling results based on the bias-corrected historical climate data from GCMs can be used for further analysis of future changes in irrigation and its impacts on global water resources.

Figure 8a shows the spatial distributions of changes in multimodel ensemble mean IWD by the end of 21st century when the global mean temperature is projected to increase by up to 6°C in the RCP8.5 emission scenario. In a warming climate, future IWDs over irrigated areas are projected to increase globally due to enhanced evaporative demands, which tends to outweigh increasing precipitation amounts [Wada *et al.*, 2013], with exceptions in regions over parts of Northeastern China, central Russian, and North Great Plain of United States. Increase in IWDs lower

Table 4. Multimodel Ensemble Mean Fraction (%) of the IWD Which Should be Met by Elsewhere (Including Water Diversions From Neighboring Basins) for the Six River Basins in the Historical and Future Periods by the IRRIG and PUMP

River Basin	Historical Period (1971–2000)		Future Period (2070–2099)	
	IRRIG	PUMP	IRRIG	PUMP
Yellow River	5.19	3.88	8.42	5.93
Haihe River	9.31	4.27	13.23	8.27
Mississippi River	1.40	0	1.98	0
Colorado River	7.74	5.23	11.68	9.71
Indus River	9.63	4.06	12.09	8.54
Ganges River	6.54	3.29	10.33	6.82

additional 10% of IWD due to increased evaporative demand in the future cannot be supported by local water resources compared to that in the current period.

To further understand the potential condition of water scarcity at the regional level, we summarize the fractions of IWD which should be met by elsewhere for the six river basins in Table 4, including but not limited to water diversions from neighboring basins, in historical and future periods under both IRRIG and PUMP scenarios. For the historical period, less than 10% of IWD need to be met potentially by water diversions in IRRIG and the fraction become smaller in PUMP in which groundwater resources were used. In the future, the picture was more complex due to the altered spatiotemporal regional distributions of surface/subsurface water supply and IWD. In general, the fraction was projected to increase in both IRRIG and PUMP in the future compared to the history. However, it should be noted that allocations of water resources within the basin is not only for irrigation but also for domestic and industrial water use, which are not accounted for in this study. Hence, the water availability within the basin for irrigation is overestimated and our estimations of the water diversions would be underestimated in this regard, which would be further complicated by other aspects of uncertainties. More detailed parameterizations of the irrigation water withdrawal processes and observations are required in order to model the water diversions in a more explicit way in the future work. We stress that the numbers provided in Table 4 should not be treated as predictions due to uncertainties discussed in section 4.

3.4. Future Impacts of Climate Change and Irrigation

Figure 9 shows the CDF of regionally averaged SW constructed from the current period and future scenarios for each GCM. The results are mixed depending on the specific GCM with regional variations. In basins such as Yellow River, Haihe River, and Ganges, climate change leads to increase in SW under a natural condition as projected by CLM driven by most GCMs in this study. With enhanced evaporative demand due to warming, irrigation amounts increase significantly and lead to potential decrease in SW. That is, the substantial increase of irrigation amounts could consume all the surplus induced by climate change and lead to subsequent decrease in SW in the IRRIG scenario when compared to the historical natural conditions. In extreme cases in Haihe River and Indus River basins, the basin-averaged surface water and local GW cannot sustain the increased irrigation demand, indicating that additional water needs to be diverted from outside the river basin in the IRRIG with climate warming. For example, vast volumes of water are transferred through the ongoing South-North water diversion project to augment the water supply in North China, which should be incorporated into water resources management models in the future. For the PUMP scenario, in contrast, the combined effects of climate change and irrigation could potentially amplify the increase of SW induced by climate change alone. For example, increases in SW induced by climate change alone are projected be around 14%, 10%, and 22% in Yellow River, Haihe River, and Ganges River basins, respectively, whereas the corresponding values are 18%, 60%, and 26% as projected by PUMP, which combines the effects of climate change and irrigation with pumping (Figure 10). This is mainly due to the effects of irrigation return flow, part of which becomes additional surface runoff. For Colorado River and Indus River basins, decreases in surface water by 21% and 10% induced by climate change alone are projected in CTRL_{ISI-MIP}. Such decreases are further aggravated when irrigation is accounted for as in both IRRIG_{ISI-MIP} and PUMP_{ISI-MIP}. However, the magnitude of SW decrease is relatively moderate in the PUMP_{ISI-MIP} scenario compared to the IRRIG scenario. For example, decreases in SW by −48% and −60% are projected in IRRIG_{ISI-MIP} while the corresponding values are −40% and −35% in PUMP_{ISI-MIP} for the Colorado and Indus River basins, respectively.

Northwest China, LWSI values could be up to 0.4, indicating that 40% of IWDs cannot be met by local SW and GW. This situation is predicted to become more severe in the future as projected by the multimodel ensemble mean results. For example, in regions of Mediterranean, Pakistan, Northern India, and southern Europe, the LWSI is predicted to increase by more than 0.1, suggesting that an

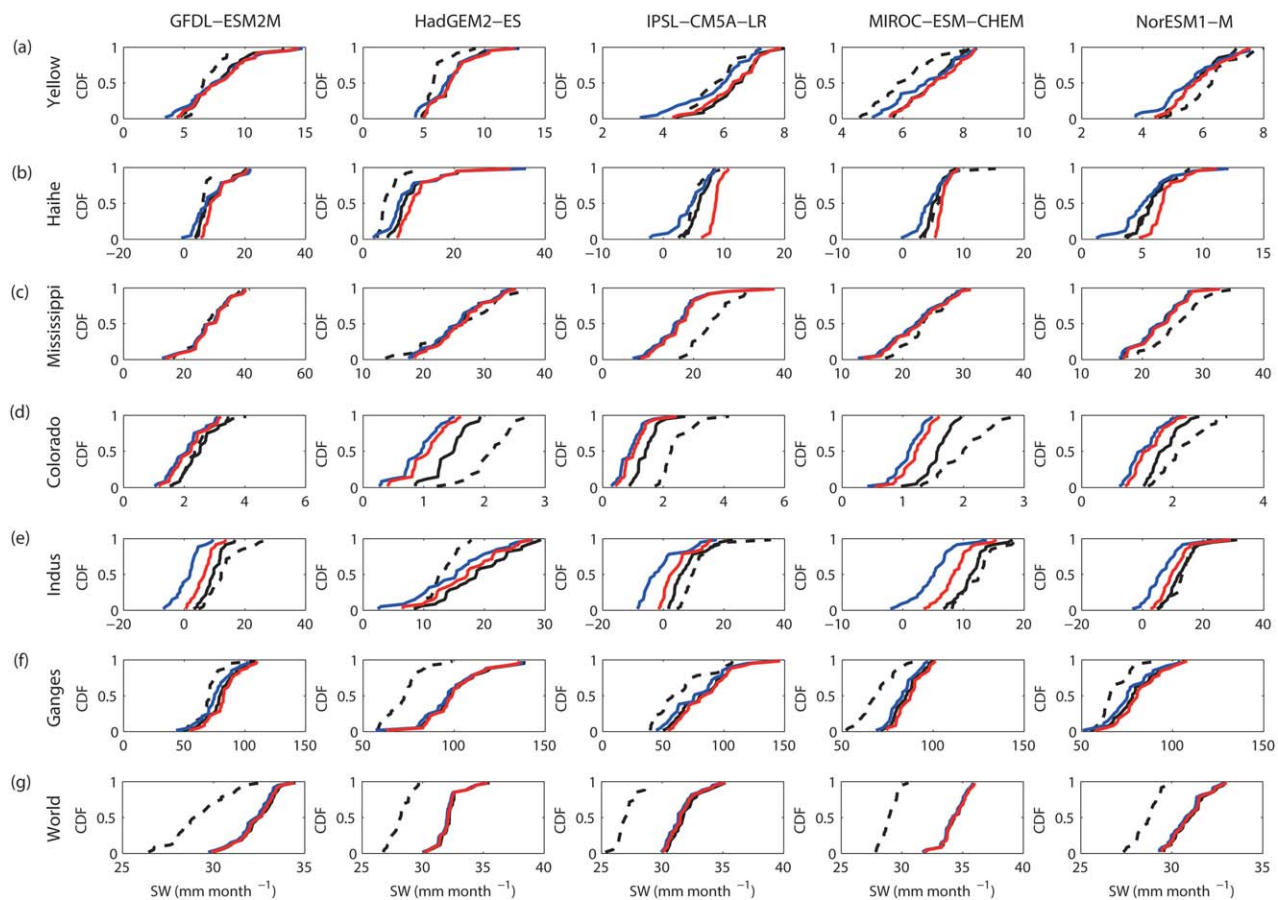


Figure 9. Cumulative distribution function (CDF) for surface water (SW, mm month^{-1}) based on 5 GCMs. The black dash line is the historical simulations by the CTRL_{ISI-MIP}, the black, blue, and red solid lines are for the future scenario by CTRL_{ISI-MIP}, IRRIG_{ISI-MIP}, and PUMP_{ISI-MIP}, respectively. Note that negative values for Haihe River and Indus River in IRRIG scenario indicate that the basin-averaged surface water resources cannot sustain the increased irrigation demand and the deficits need to be met by water diversion from outside the river basin.

When averaged globally, the combined impacts of climate change and irrigation with pumping are projected to result in an increase in SW by about 14%, similar to the magnitude of that induced by climate change.

Similarly, a mixture of regional GW increase/decrease induced by climate change alone is projected in the future (Figure 11). In Yellow River and Colorado River basins, climate change could lead to decreases in GW as projected by the CLM simulations driven by most GCMs. Conversely, in Mississippi River basin, increase in GW induced by climate change alone is predicted. For Haihe River, Indus River, and Ganges River basins, negligible changes of GW are predicted. When irrigation water use is accounted for, the decrease in GW due solely to climate change is alleviated through the additional recharge by irrigation return flow in IRRIG_{ISI-MIP} or aggravated due to GW abstraction in PUMP_{ISI-MIP} with region-specific magnitudes. For river basins relying heavily on GW for irrigation, the combined effects of climate change and irrigation are projected to result in much larger decreases in GW than the sole climate change impact. For example, in Haihe, Indus, and Ganges River basins where climate change resulted in slight increase in GW, excessive extraction for irrigation has the potential to exacerbate the problem in these regions with GW decreasing by 7%, 7%, and 6% respectively (Figure 12). Averaged over the world, however, the decrease in GW induced by climate change alone is reversed when irrigation is accounted for in IRRIG_{ISI-MIP}, while such a decrease remains relatively unchanged in PUMP_{ISI-MIP}.

4. Uncertainties and Limitations

In this study, the key parameters were required and derived from the global inventory of groundwater and surface water use for irrigation developed by Siebert *et al.* [2010] using census-based statistical data. Large uncertainties remain in the areas equipped for irrigation with groundwater and on the related water uses

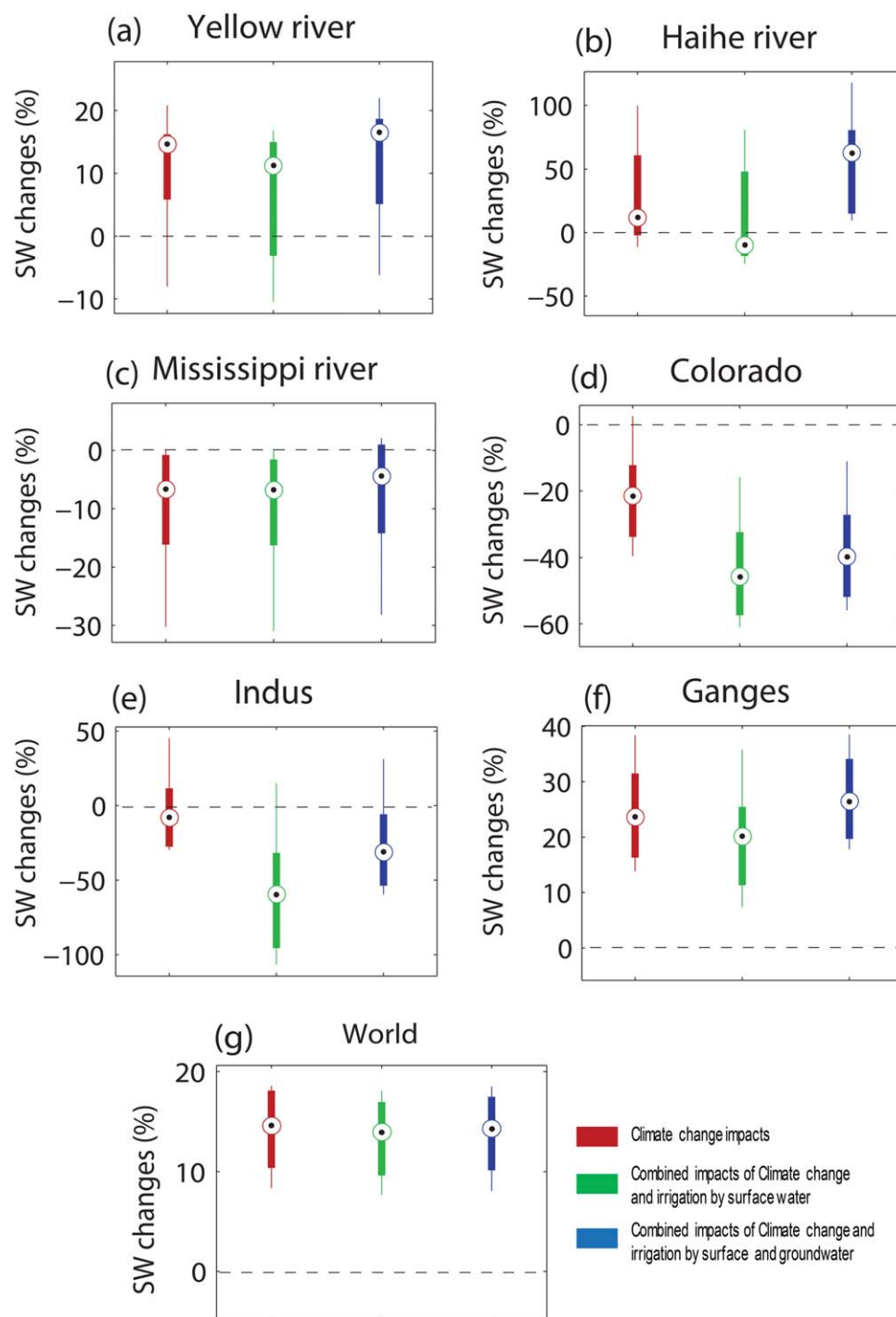


Figure 10. Box plots of relative changes in surface water by 2070–2099 relative to 1971–2000. Box edges illustrate the 25th and 75th percentile, the central mark is the median and whiskers mark the 5th and 95th percentiles. Red bar is the impacts of sole climate change. Green bar is the combined effects of climate change and irrigation with surface water. Blue bar is the combined effects of climate change and irrigation with surface water and groundwater. All climate and combined effects are compared with the control period naturalized simulations.

due to use of various definitions for irrigated land and its water sources and issues such as missing data [Siebert *et al.*, 2010; Döll *et al.*, 2012]. Large uncertainties from the data sets would propagate through the model and affect the results. Specifically, F_{irrig} is the parameter governing the estimated irrigation water demand, which is met by withdrawing surface water and groundwater based on the ratio F_{grd} . The two

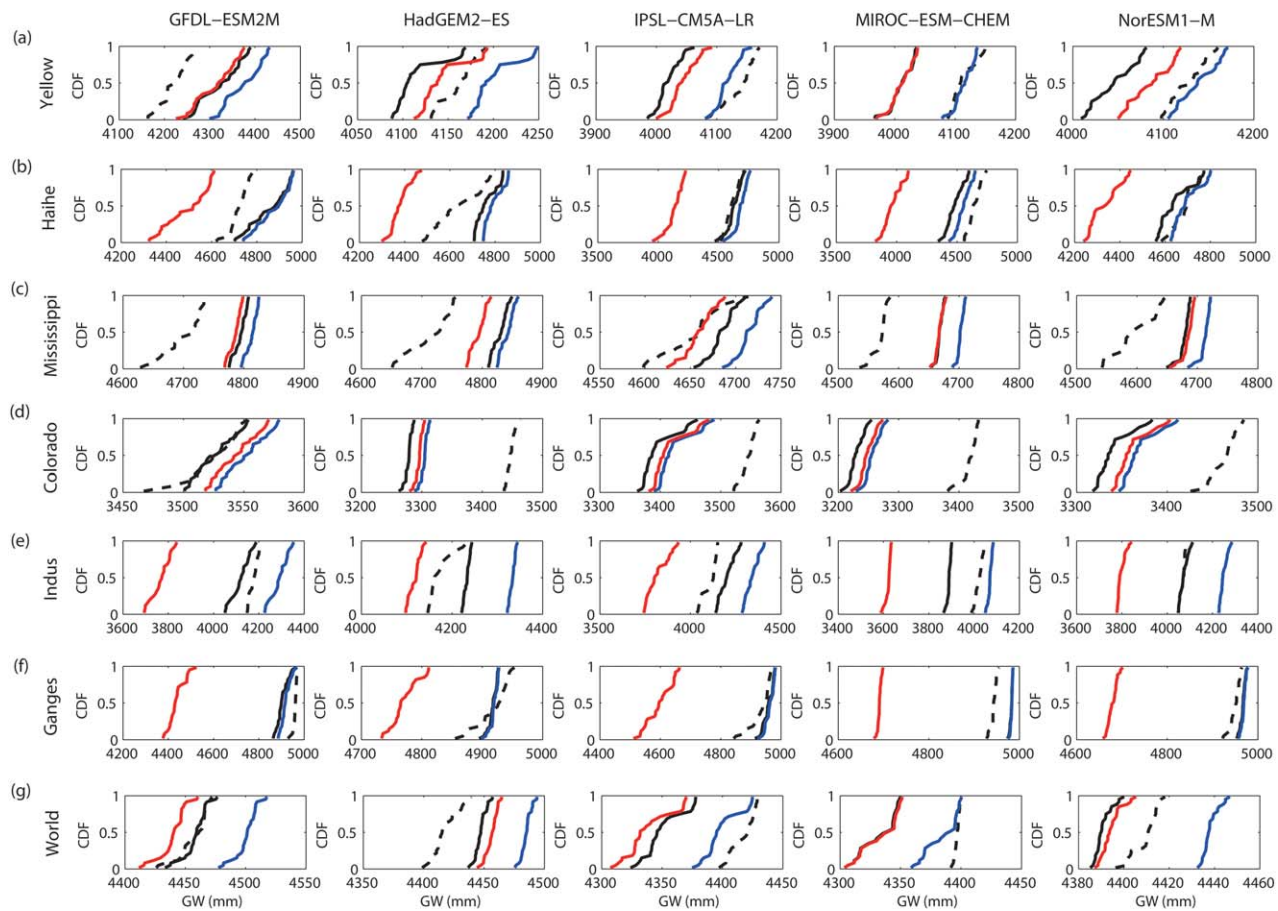


Figure 11. Same as Figure 9 but for groundwater.

parameters, among others, can have large influences on simulations of SW/GW changes. Quantitative evaluations of parameter sensitivity require additional analyses, similar to those performed in our previous study [Leng *et al.*, 2013], which is out of the scope of this study. Although the total irrigation amounts as simulated by the calibrated model can capture the interannual variability and trends well when compared to census data [Leng *et al.*, 2013; Leng and Tang, 2014], the temporally constant fraction of groundwater irrigation to total irrigation may dampen the seasonal variability and trends of surface water and groundwater conditions. We acknowledge that this is an issue to be addressed the future, potentially through coupling with an integrated assessment model, as demonstrated in Voisin *et al.* [2013b].

Deficiencies of the irrigation scheme in CLM4 should also be acknowledged. Such deficiencies, mainly due to data availability at the global scale, are dealt with through a number of assumptions. In the IRRIG scenario, the irrigation demand was met only by surface water, which neglected the groundwater source (compared to PUMP scenario). Even with additional water from groundwater in the PUMP scenario, it is not necessarily true that the demand could be met without considering reservoir storage and release, water diversions, and other sources. It is therefore assumed that the unmet irrigation demand can be delivered from the surface water from other grid cells within the basin in this study. However, such an assumption is not valid for transboundary river basins where multiple countries within one basin do not often share their water resources in an optimal matter. Since the exact location of grid cells contributing to the irrigation water cannot be well identified, we are more confident about the results averaged at the river basin scale (i.e., six basins are selected for analysis), which is more relevant to the water management. When the basin-average water resource is not enough to meet the demand, water extraction from places outside the basin is assumed to represent a simplification of reality. Therefore, from above discussions, the irrigation amount is probably overestimated and should be regarded with caution. Explicit representations of sources for water withdrawals should be made in our future efforts on model improvements [Voisin *et al.*, 2013a,

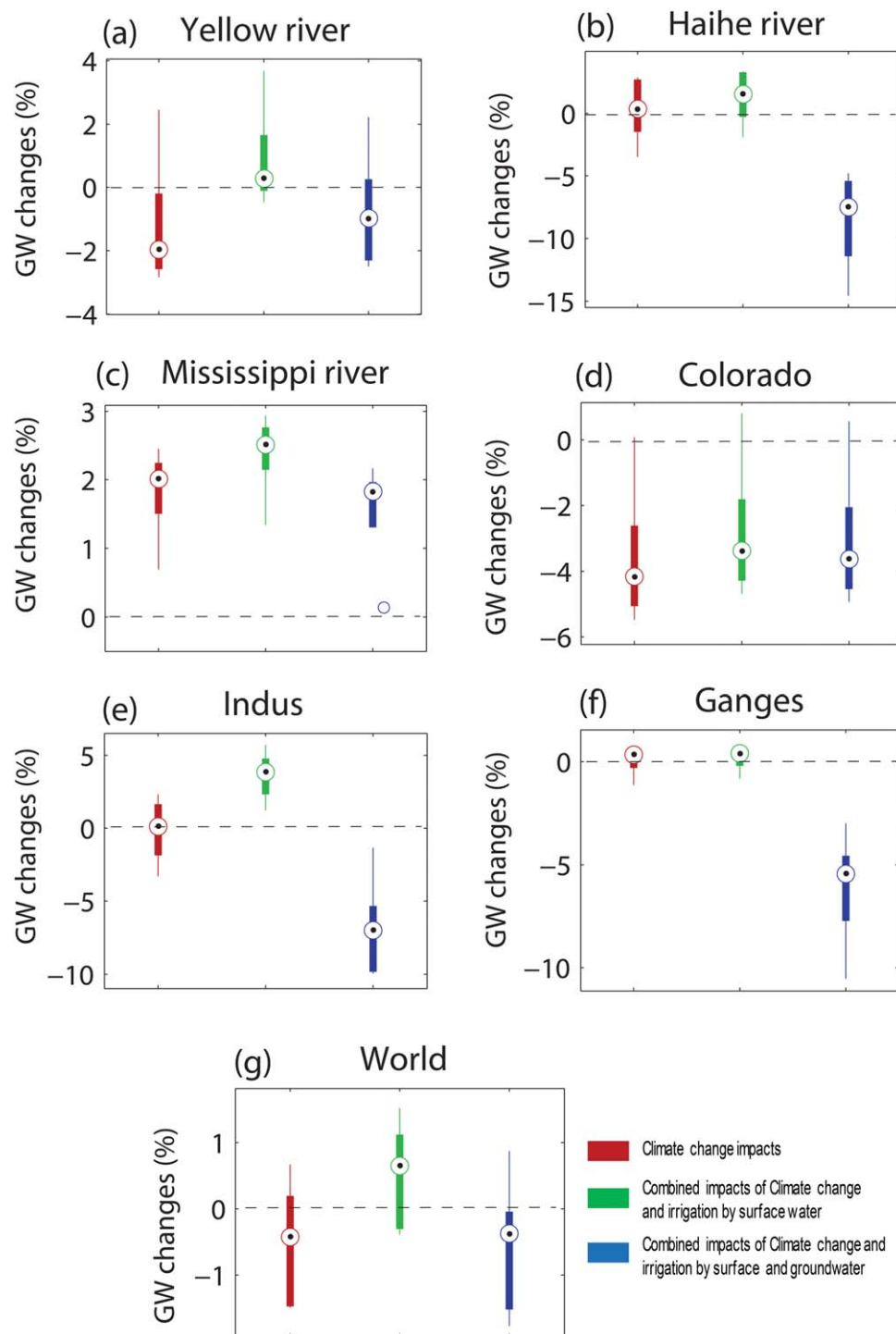


Figure 12. Same as Figure 10 but for groundwater.

2013b]. At the same time, our irrigation water use is estimated as consumptive water use that ignores losses due to drainage or evaporation during delivery [Qian *et al.*, 2013; Leng and Tang, 2014]. Loss of irrigation water through percolation may also result in low irrigation efficiency. This is particularly true when groundwater pumping is considered in water extractions as well demonstrated and discussed in our previous work [Leng *et al.*, 2014a]. Nevertheless, it is important to note that the reference data by Siebert *et al.* [2010] were estimated based on the optimal demand for crop growth, which are used as the target for our model

calibration. Therefore, our simulated irrigation amounts would to certain extent be overestimated against reality. When combined together, the overestimation and underestimation of irrigation amounts could potentially offset each other.

Some uncertainties in the relative changes in SW and GW are related to the hydrological representations in CLM4, which determines surface and subsurface runoff and soil moisture state that influence infiltration and return flow. Similar to other large-scale land surface models [e.g., Ozdogan *et al.*, 2010; Zou *et al.*, 2014; Pokhrel *et al.*, 2014], the complex interactions between river water and groundwater are not explicitly represented, which is important in regions such as the lower Mississippi where lateral flow of groundwater played a large role in the formation of shallow groundwater tables in the floodplain. Based on USGS census and historic streamflow data, Kustu *et al.* [2010] showed that GW pumping led to significant decrease in SW over the Ogallala Aquifer. The importance of GW–SW interactions was also demonstrated in high resolution modeling of the subsurface water dynamics [Ferguson and Maxwell, 2012; Condon and Maxwell, 2014]. Since this study focuses on the long-term term (i.e., 30 years) and large-scale water balance rather than the short-temporal dynamics, it is expected that the deficiency in accounting for lateral groundwater flow may not be critical.

In this study, the effects of CO₂ rises on plant growth and productivity are not considered in irrigation estimations, which may turn out to be beneficial at the global scale [Konzmann *et al.*, 2012]. Moreover, land use change was not considered and the irrigated areas were held constant in this study, because comprehensive data of land use change are not available. Hence, our results should mainly highlight the impacts of climatic changes on terrestrial hydrology and irrigation water use in current irrigated regions. Under future climate change, in particular, the scenario chosen in this study (i.e., RCP8.5), it is expected that more land is used as cropland and pasture to sustain the increasing global population. Given the predicted increase in aridity under global warming [Fu and Feng, 2014], the projected increase of crop areas would require more irrigation water and subsequently lead to larger impacts of irrigation. Recognizing such needs, we have developed an approach to downscale land use information from the integrated assessment model, the Global Change Assessment Model (GCAM), for land use land cover scenarios consistent with the RCP or SSP scenarios and could be used as inputs to impact assessment models [West *et al.*, 2014]. As part of our ongoing research, we have extended the approach in West *et al.* [2014] to include downscaling of irrigated areas over the conterminous U.S. The extension of the downscaling approach globally is also ongoing and we will report progress when results are available. Moreover, human water use consists of not only irrigation, but also industrial and domestic water consumption. All water demand if partially or totally met could affect the natural land surface hydrology [Zou *et al.*, 2014; Voisin *et al.*, 2013b]. Without incorporation of industrial and domestic water consumption, the surface water and groundwater changes may be underestimated. Thus, our findings are limited to how irrigation in the present irrigation areas responds to climate change directly through climate change effects on the water balance and indirectly through changes in evaporative demand that influence irrigation water demand. For a more complete picture, future studies should take into account for land use change scenarios [West *et al.*, 2014], effects of specific crop types and crop growth [Chen and Xie, 2011, 2012], and changes in irrigation efficiencies. Both climate change and irrigation could alter the seasonal water balance, which should also be investigated in the future, as it has important implications to managing water resources. Finally, climate change impacts on irrigation and the direct/indirect impacts on surface/subsurface water should depend on the selected emission scenarios, climate models, downscaling approaches, time periods, etc. [Leng *et al.*, 2015].

5. Summary and Conclusions

Global change encompasses spatial-temporal changes in the characteristics of climate variables, and derived changes in terrestrial processes, including human water use that affects surface and subsurface hydrology. Potential effects of climate change combined with irrigation water use on surface waters have been studied recently [e.g., Haddeland *et al.*, 2014; Forzieri *et al.*, 2014]. However, studies on how climate and irrigation might jointly influence the groundwater systems are very limited, especially at global scale. In this paper, we extended the groundwater pumping scheme implemented in CLM4 by Leng *et al.* [2014a] to global scale. The simulated irrigation amounts were first calibrated against the global inventory developed by Siebert *et al.* [2010] and then partitioned into surface water and groundwater withdrawals based on the

ratio archived in the same global inventory. Then, the impacts of climate on surface water and ground water through natural and human-induced processes were investigated.

The results indicate that the impact of irrigation water withdrawals on the long-term annual mean global water resources is small. However, in some river basins, the impacts of irrigation are significant with two major opposing effects on surface water (SW)/groundwater (GW): SW depletion/GW accumulation in regions with irrigation primarily fed by SW, and SW accumulation/GW depletion in regions with irrigation fed primarily by groundwater. Furthermore, irrigation depending on SW tends to have much larger impacts on the low-flow conditions of surface water compared to that on the high-flow conditions, suggesting that intensive water use can further exacerbate low-flow conditions in watersheds, increasing their vulnerability to drought. At the regional scale, SW and GW could be increasing or decreasing by the end of the 21st century, when climate change is the only factor to be considered. However, given that irrigation demand is projected to increase significantly for irrigated areas across the world due to enhanced evaporative demand, the net effect of water supply and demand changes could lead to a growing water scarcity for irrigation especially in major irrigated river basins. The impact of climate change alone on global surface water and groundwater would be further aggravated/alleviated by the increased irrigation with climate warming in the future.

Despite the limitations as discussed in section 5, this study represents a significant effort toward using CLM in an earth system modeling framework to simulate irrigation and demonstrate their impacts on global surface water and groundwater resources. Its novelty lies in its explicit account for the sources of irrigation water and their effects on both surface water and groundwater resources in the context of future climate change, which turned out to be two opposing effects depending on the irrigation source and study region. Our results emphasize the importance of accounting for the sources for irrigation in climate change impact assessment studies, based on which mitigation and adaptation strategies are designed.

Acknowledgments

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